Machine Learning

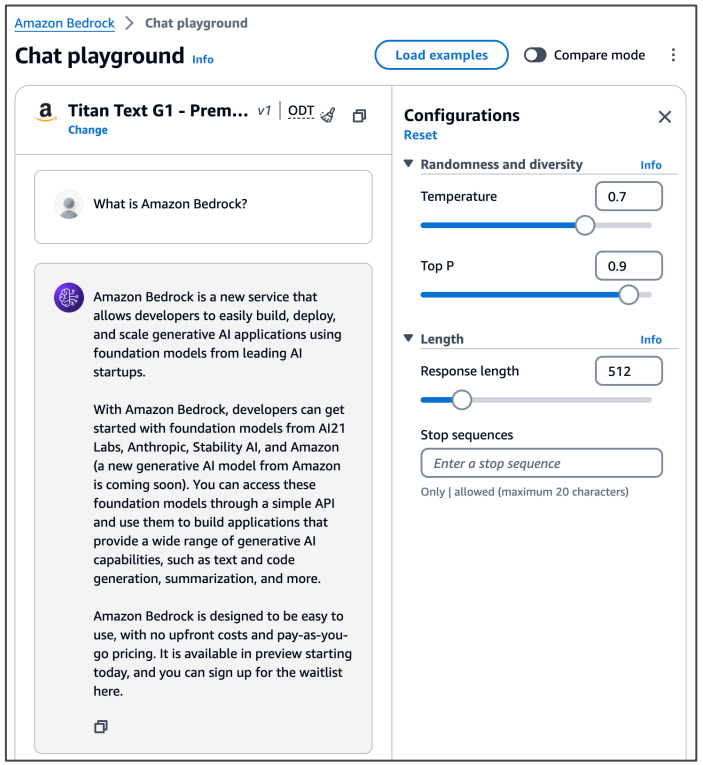
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock



- Build Generative AI (Gen-AI) applications on AWS
- Fully-managed service, no servers for you to manage
- Keep control of your data used to train the model
- Pay-per-use pricing model
- Unified APIs
- Leverage a wide array of foundation models
- Out-of-the box features: RAG, LLM Agents…
- Security, Privacy, Governance and Responsible AI features



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock – Foundation Models

- Access to a wide range of Foundation Models (FM)













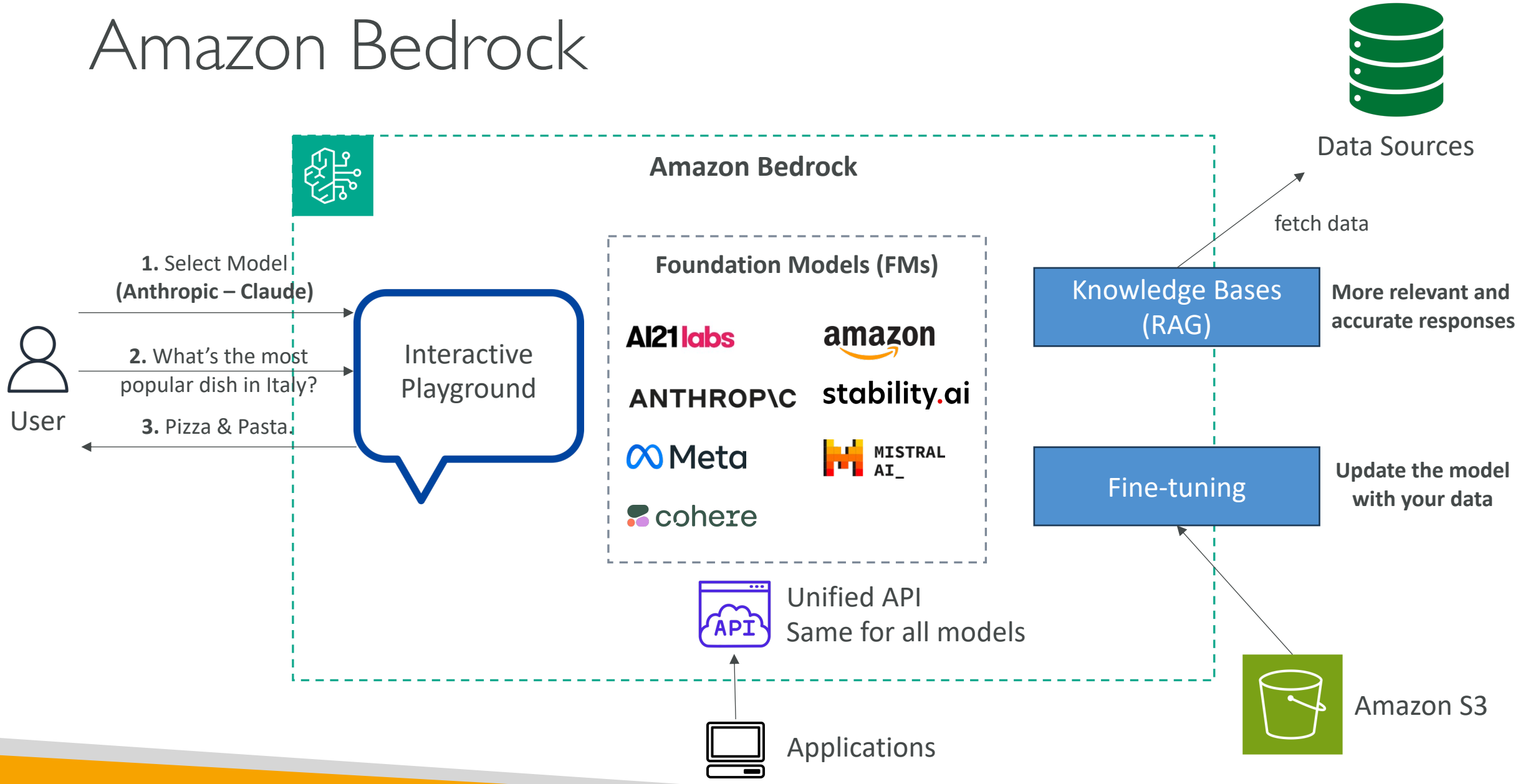


- Amazon Bedrock makes a copy of the FM, available only to you, which you can further fine-tune with your own data
- None of your data is used to train the FM

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Bedrock



NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon Rekognition



- Find **objects, people, text, scenes** in **images and videos** using ML
- **Facial analysis** and **facial search** to do user verification, people counting
- Create a database of "familiar faces" or compare against celebrities
- Use cases:
  - Labeling
  - Content Moderation
  - Text Detection
  - Face Detection and Analysis (gender, age range, emotions…)
  - Face Search and Verification
  - Celebrity Recognition
  - Pathing (ex: for sports game analysis)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Rekognition – Content Moderation

- Detect content that is inappropriate, unwanted, or offensive (image and videos)
- Used in social media, broadcast media, advertising, and e-commerce situations to create a safer user experience
- **Set a Minimum Confidence Threshold for items that will be flagged**
- Flag sensitive content for manual review in Amazon Augmented AI (A2I)
- Help comply with regulations



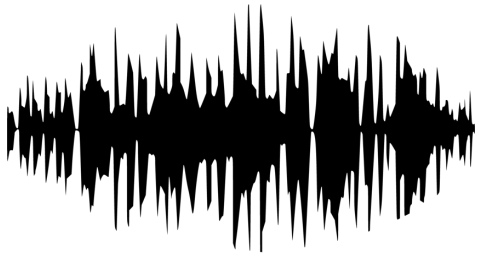
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Transcribe



- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition** (ASR) to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification for multi-lingual audio**
- Use cases:
  - transcribe customer service calls
  - automate closed captioning and subtitling
  - generate metadata for media assets to create a fully searchable archive



"Hello my name is Stéphane.
I hope you're enjoying the course!

© Stephane Maarek

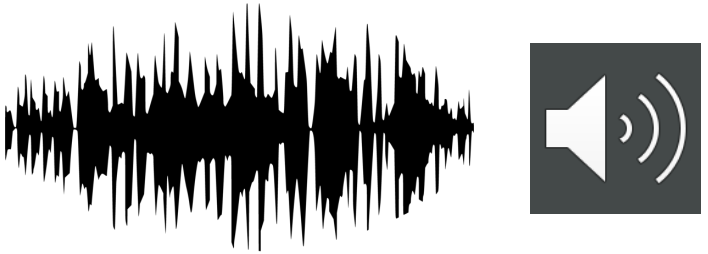
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Polly



- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk

Hi! My name is Stéphane
and this is a demo of Amazon Polly



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Polly – Lexicon & SSML

- Customize the pronunciation of words with **Pronunciation lexicons**
  - Stylized words: St3ph4ne => "Stephane"
  - Acronyms: AWS => "Amazon Web Services"
- Upload the lexicons and use them in the **SynthesizeSpeech** operation

- Generate speech from plain text or from documents marked up with **Speech Synthesis Markup Language (SSML)** – enables more customization
  - emphasizing specific words or phrases
  - using phonetic pronunciation
  - including breathing sounds, whispering
  - using the Newscaster speaking style

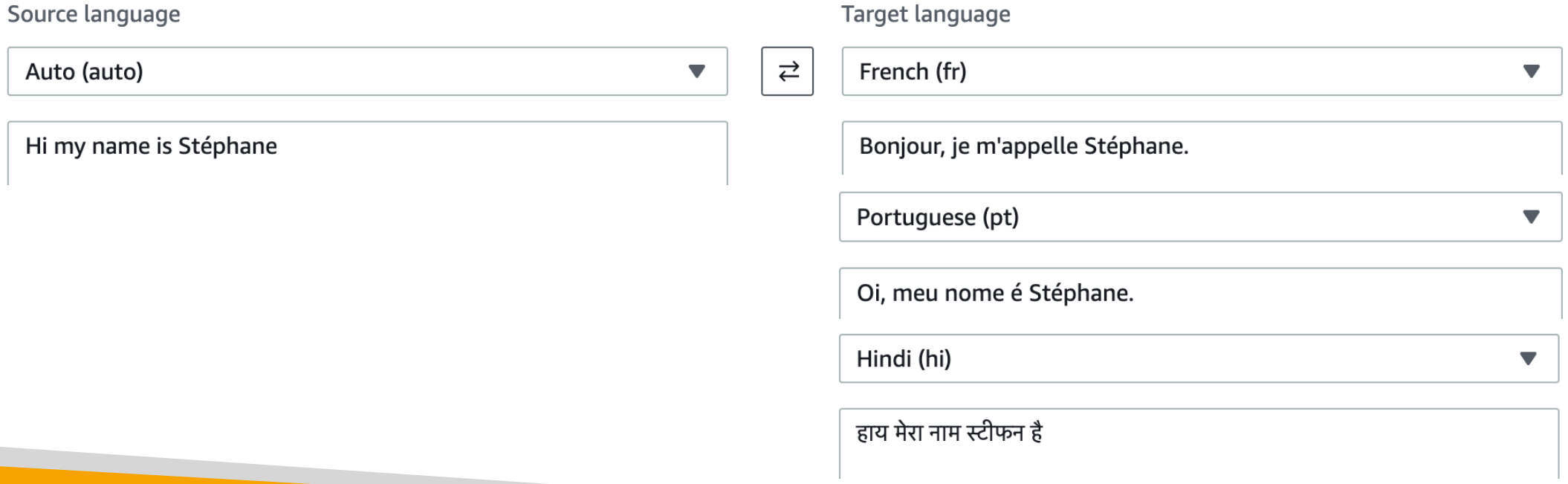
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Translate



- Natural and accurate **language translation**
- Amazon Translate allows you to **localize content** - such as websites and applications - for **international users**, and to easily translate large volumes of text efficiently.

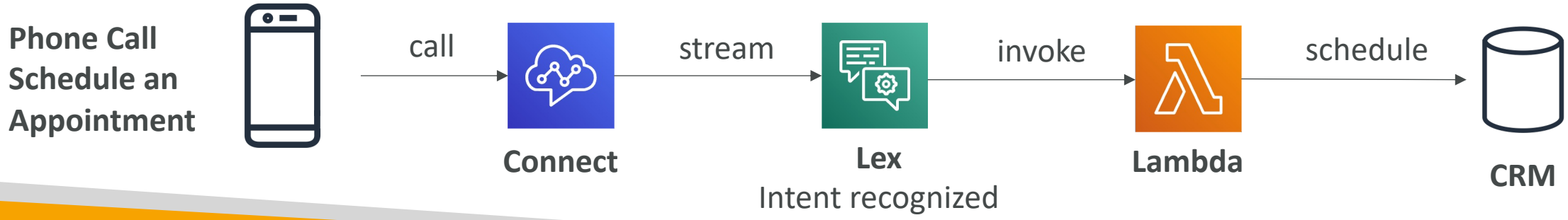


© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Lex & Connect

- **Amazon Lex:** (same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural Language Understanding to recognize the intent of text, callers
  - Helps build chatbots, call center bots
- **Amazon Connect:**
  - Receive calls, create contact flows, cloud-based **virtual contact center**
  - Can integrate with other CRM systems or AWS
  - No upfront payments, 80% cheaper than traditional contact center solutions



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Comprehend



- For **Natural Language Processing – NLP**
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
  - Language of the text
  - Extracts key phrases, places, people, brands, or events
  - Understands how positive or negative the text is
  - Analyzes text using tokenization and parts of speech
  - Automatically organizes a collection of text files by topic
- Sample use cases:
  - analyze customer interactions (emails) to find what leads to a positive or negative experience
  - Create and groups articles by topics that Comprehend will uncover

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Comprehend Medical



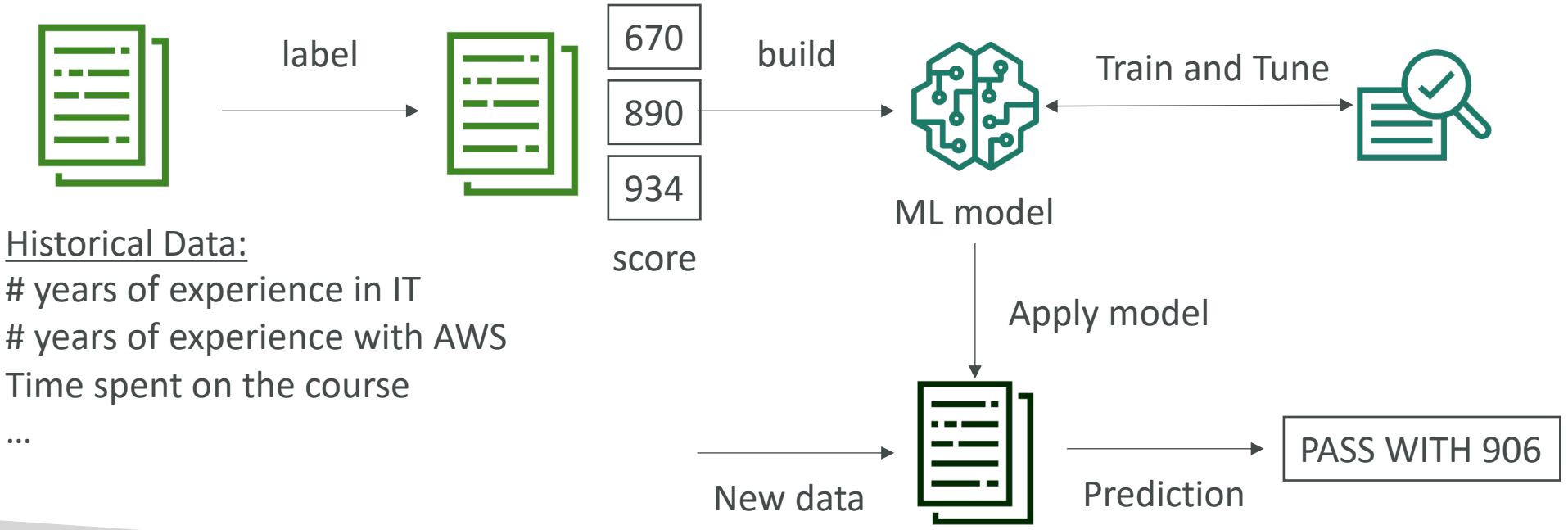
- Amazon Comprehend Medical detects and returns useful information in unstructured clinical text:
  - Physician's notes
  - Discharge summaries
  - Test results
  - Case notes
- **Uses NLP to detect Protected Health Information (PHI)** – *DetectPHI* API
- Store your documents in Amazon S3, analyze real-time data with Kinesis Data Firehose, or use Amazon Transcribe to transcribe patient narratives into text that can be analyzed by Amazon Comprehend Medical.

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon SageMaker AI

- Fully managed service for developers / data scientists to build ML models
- Typically, difficult to do all the processes in one place + provision servers
- Machine learning process (simplified): predicting your exam score



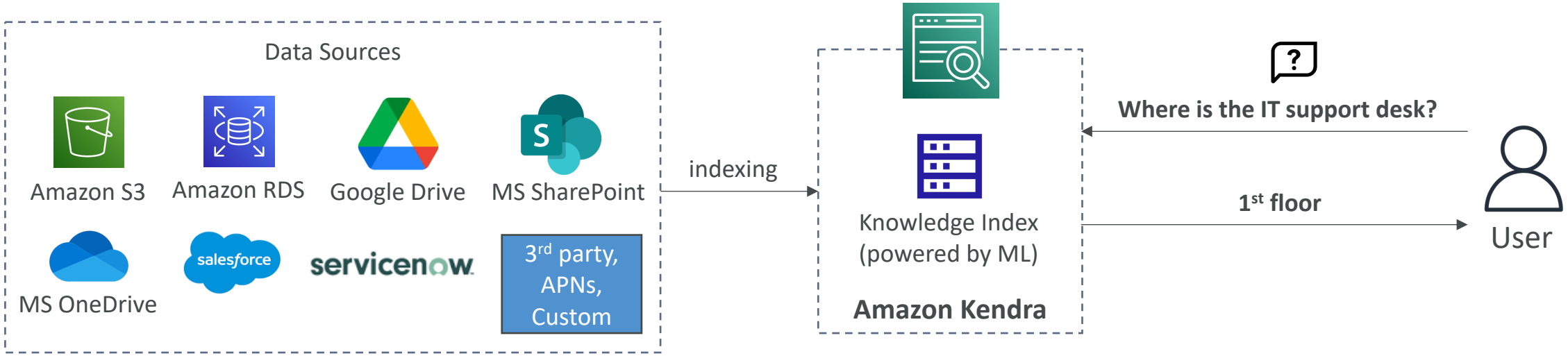
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Kendra



- Fully managed **document search service** powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, MS Word, FAQs…)
- Natural language search capabilities
- Learn from user interactions/feedback to promote preferred results **(Incremental Learning)**
- Ability to manually fine-tune search results (importance of data, freshness, custom, …)



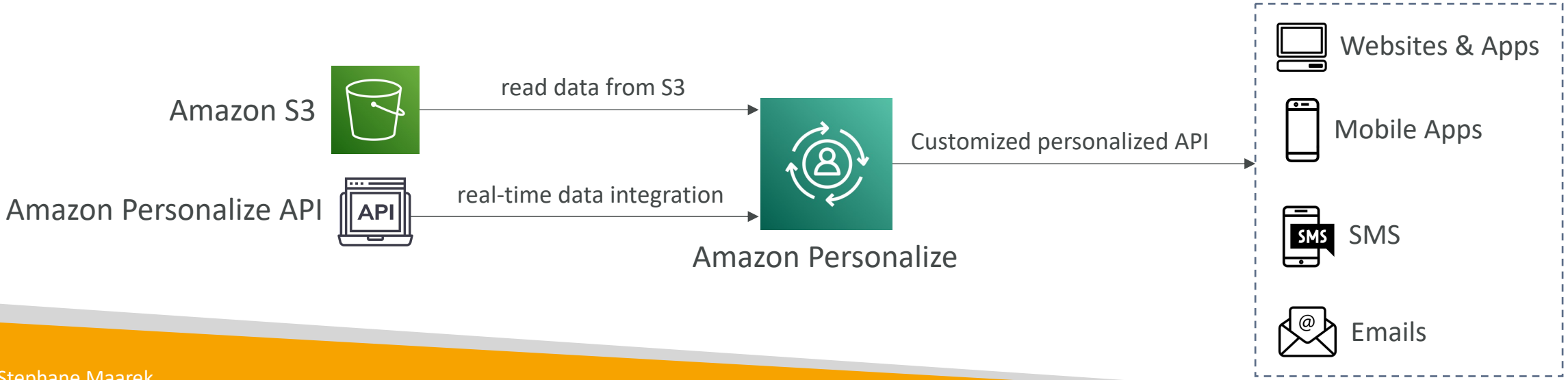
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Personalize



- Fully managed ML-service to build apps with real-time personalized recommendations
- Example: personalized product recommendations/re-ranking, customized direct marketing
  - Example: User bought gardening tools, provide recommendations on the next one to buy
- Same technology used by Amazon.com
- Integrates into existing websites, applications, SMS, email marketing systems, …
- Implement in days, not months (you don't need to build, train, and deploy ML solutions)
- Use cases: retail stores, media and entertainment…



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon Textract



- Automatically extracts text, handwriting, and data from any scanned documents using AI and ML



analyze

Amazon Textract

result

```
{
   "Document ID": "123456789-005",
   "Name": "",
   "SEX": "F",
   "DOB": "23.05.1997",
   ...
}
```

- Extract data from forms and tables
- Read and process any type of document (PDFs, images, …)
- Use cases:
  - Financial Services (e.g., invoices, financial reports)
  - Healthcare (e.g., medical records, insurance claims)
  - Public Sector (e.g., tax forms, ID documents, passports)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS Machine Learning - Summary

- **Rekognition:** face detection, labeling, celebrity recognition
- **Transcribe:** audio to text (ex: subtitles)
- **Polly:** text to audio
- **Translate:** translations
- **Lex:** build conversational bots – chatbots
- **Connect:** cloud contact center
- **Comprehend:** natural language processing
- **SageMaker:** machine learning for every developer and data scientist
- **Kendra:** ML-powered search engine
- **Personalize:** real-time personalized recommendations
- **Textract:** detect text and data in documents

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com